

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – SHORT ANSWER TEST

Course Code:	CSA1613	Course Title:	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	10% of CIA	Total Marks:	100 (5 Questions × 20 Marks)
CO1 Statement:	Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2		
Student Name:	Katappagari Dhoni	Reg. No.:	192411140
Section / Batch:	CSE	Date:	15/07/2026

Instructions to Students

- Answer all questions. Each question carries 20 Marks. Total: 100 Marks.
- Clearly show all requirements, process, operations.
- Use appropriate models, schema represented scenario wise
- Justify each step in different dimension specified and measures clearly.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Short Answer Test	Architecture of Data Warehouse, ETL, OLAP	Q1 – Q5	Q1 –20 Q2 –20 Q3 –20 Q4 –20 Q5 –20
GRAND TOTAL:			100 Marks

SECTION A – SHORT ANSWER TEST

1. Suppose that a data warehouse for City Hospital consists of the following four dimensions: patient, doctor, department, and time, and two measures visit_count and avg_charge. At the lowest conceptual level, avg_charge stores the actual consultation fee charged for a patient visit. At higher conceptual levels, it stores the average fee.

- Draw a snowflake schema diagram for the data warehouse.
- Starting with the base cuboid [patient, doctor, department, time], what specific OLAP operations should be performed to list the average consultation fee department-wise for each year?
- If each dimension has four levels (including all), how many cuboids will this cube contain (including base and apex cuboids)?

2. A smart city project is collecting data from multiple sources, including:

- Traffic sensors for vehicle counts and congestion levels.
- IoT devices for monitoring air quality, noise levels, and energy consumption.
- Public transportation systems for ridership data and punctuality.
- Citizen feedback from a mobile app.

Questions:

Fact and Dimension Design:

- Design a star schema to analyze traffic congestion patterns, energy consumption trends, and public transportation efficiency.
- What measures and dimensions would you include? Justify your design.

3. Suppose that a banking data warehouse contains two fact tables: Transaction_Fact and Loan_Fact, sharing dimensions customer, branch, time, and account_type.

- Transaction_Fact → transaction_count, amount
- Loan_Fact → loan_amount, interest

(a) Draw a galaxy schema diagram.

(b) Identify OLAP operations to obtain branch-wise average transaction amount and loan amount yearly.

(c) If each dimension has four levels, calculate total cuboids.

4. A manufacturing company monitors product quality and production efficiency. The data warehouse contains:

• Dimensions:

Product: Category, type, and batch.

Factory: Location, department, and line number.

Time: Year, quarter, month, and week.

Supplier: Region, rating, and contract type.

• Measures:

Defect rate.

Production units.

Supplier reliability percentage.

Questions:

- Roll-up: Summarize defect rates by product category for the current year across all factories.
- Drill-down: Focus on a specific factory to analyze defect rates by department and product type.
- Slice: Isolate production unit data for suppliers with a rating of 4+ across all regions.
- Dice: Analyze defect rates and supplier reliability for selected product categories and factory locations.

5. A retail chain manages inventory and sales data across multiple stores. The data warehouse contains:

• Dimensions:

Product: Category, subcategory, and SKU.

Store: Region, city, and store ID.

Time: Year, quarter, month, and day.

Customer: Membership tier, age group, and gender.

• Measures:

Inventory turnover.

Total sales revenue.

Units sold.

Questions:

1. Roll-up: Summarize inventory turnover by product category across all stores for the last quarter.
2. Drill-down: Focus on sales data for electronics in a specific region, analysing by subcategory and SKU.
3. Slice: Isolate sales revenue data for male customers in the Bronze membership tier across all stores.
4. Dice: Analyze sales revenue and units sold for selected cities and product categories during specific time periods.

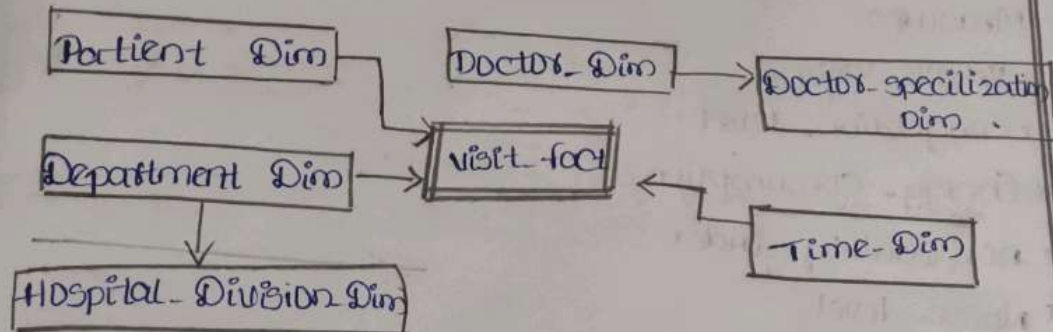
Code of Conduct

I, K. Dhawan (192611140) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Dhawan

City Hospital Dataware house

a) Snowflake Schema



b) OLAP Operations

1. Rollup Customer $\rightarrow$ All
2. Rollup Account Type $\rightarrow$ All
3. Rollup Time $\rightarrow$ Year
4. Keep Branch

Result Cuboid:

[All, Branch, Year, All]

Aggregate:

- * AVG (Amount)
- * AVG (Loan-Amount)

c) Number of Cuboids

Each dimension has 4 levels.

$$4 \times 4 \times 4 \times 4 = 256$$

Total cuboids = 256

2. Smart City Project:

- fact table: smartcity-fact

Measures:

- * Vehicle count
- * Congestion level
- * Energy consumption
- * Air Quality Index
- * Noise level
- * Ridership
- * Punctuality
- * Feedback score

Dimensions:

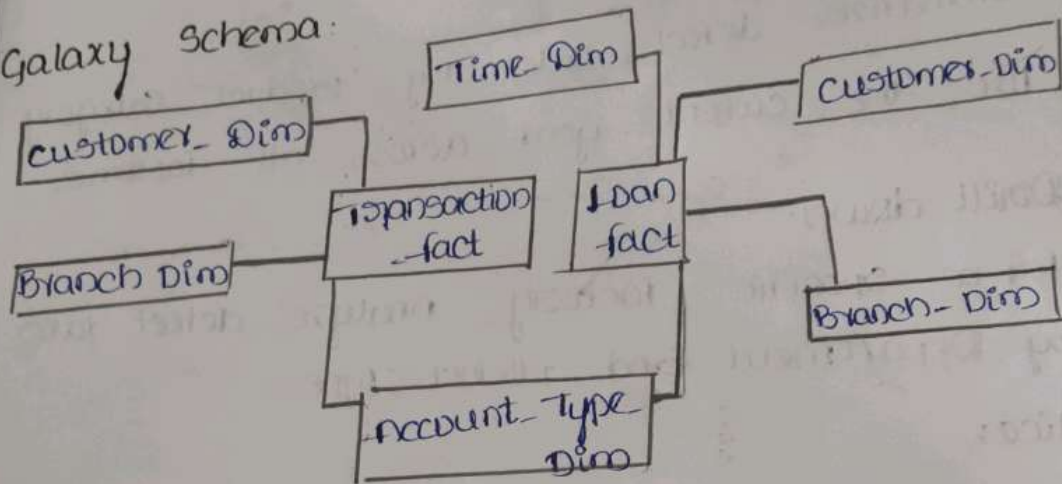
- * Time Dim (Day, Month, Year)
- * Location Dim (Area, zone, city)
- * Transport Dim (Route, vehicle Type)
- * Citizen Dim (Age group, feedback Type)

Justification:

Measures store numerical data for analysis, while dimensions provide different perspectives such as time, location, transport, and citizen feedback.

Banking Data Warehouse

a) Galaxy Schema:



b) OLAP Operations:

- * Roll up customer $\rightarrow$ ALL
- * Roll up Account Type $\rightarrow$ ALL
- * Roll up Time $\rightarrow$ Year
- * Keep Branch

Result:

[ALL, ALL, Year, Branch]

Aggregate:

- * AVG (Amount)
- * AVG (Loan-Amount)

c) Total Cuboids:

$$4^4 = 256$$

Total cuboids = 256

4. Manufacturing Company

Roll up:

Summarize defect rates by product category for the current year across all factories

Drill down:

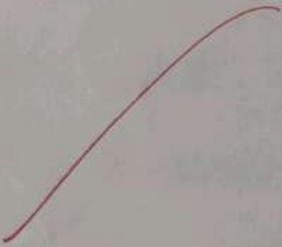
for a specific factory analyze defect rates by Department and product Type.

Slice:

Select suppliers with Rating ≥ 4 and analyze production units

Dice:

Analyze Defect Rate and Supplier Reliability for selected product categories and factory locations



Retail chain

1. Roll up:

Summarize inventory turnover by product category across all stores for the last quarter.

Operation:

- * product: SKU $\rightarrow$ Subcategory $\rightarrow$ Category
- * Store: ID $\rightarrow$ City $\rightarrow$ Region $\rightarrow$ ALL
- * Time: Day $\rightarrow$ Month Quarter

2. Drill-down

Analyze electronics sales in specific region.

Operation:

Select category = electronics

Drill down: Category $\rightarrow$ Subcategory $\rightarrow$ SKU

3. Slice:

Isolate sales revenue for male customers in Bronze membership tier

Condition:

* Gender = Male

* Membership Tier = Bronze

4. Dice
Analyze sales revenue and units sold for
selected cities and product categories during
specific periods

Conditions:

cities: chennai, Mumbai

categories: electronics, clothing

Time: year 2025

